Experiment No. 18 – Image Cropping, Copying and Pasting Using ROI

1. Aim

To implement image cropping, copying and pasting by selecting a Region of Interest (ROI) from the source image using OpenCV.

2. Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

# Read image
filename = list(uploaded.keys())[0]
image = cv2.imdecode(
    np.frombuffer(uploaded[filename], np.uint8),
    cv2.IMREAD_COLOR
)

if image is None:
    print("Error: Image not loaded")
else:
    # Get image dimensions
    h, w = image.shape[:2]
```

```
# Select Region of Interest (ROI)
x1, y1 = w//4, h//4
x2, y2 = 3*w//4, 3*h//4

# Crop the selected region
roi = image[y1:y2, x1:x2]

# Copy original image
result = image.copy()

# Paste ROI at a different location
rh, rw = roi.shape[:2]
px, py = 10, 10

result[py:py+rh, px:px+rw] = roi.copy()

# Display images
plt.figure(figsize=(12, 6))

plt.subplot(1, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
```

```
plt.imshow(cv2.cvtColor(roi, cv2.COLOR_BGR2RGB))
plt.title("Cropped ROI")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(cv2.cvtColor(result, cv2.COLOR_BGR2RGB))
plt.title("ROI Copied and Pasted")
plt.axis("off")

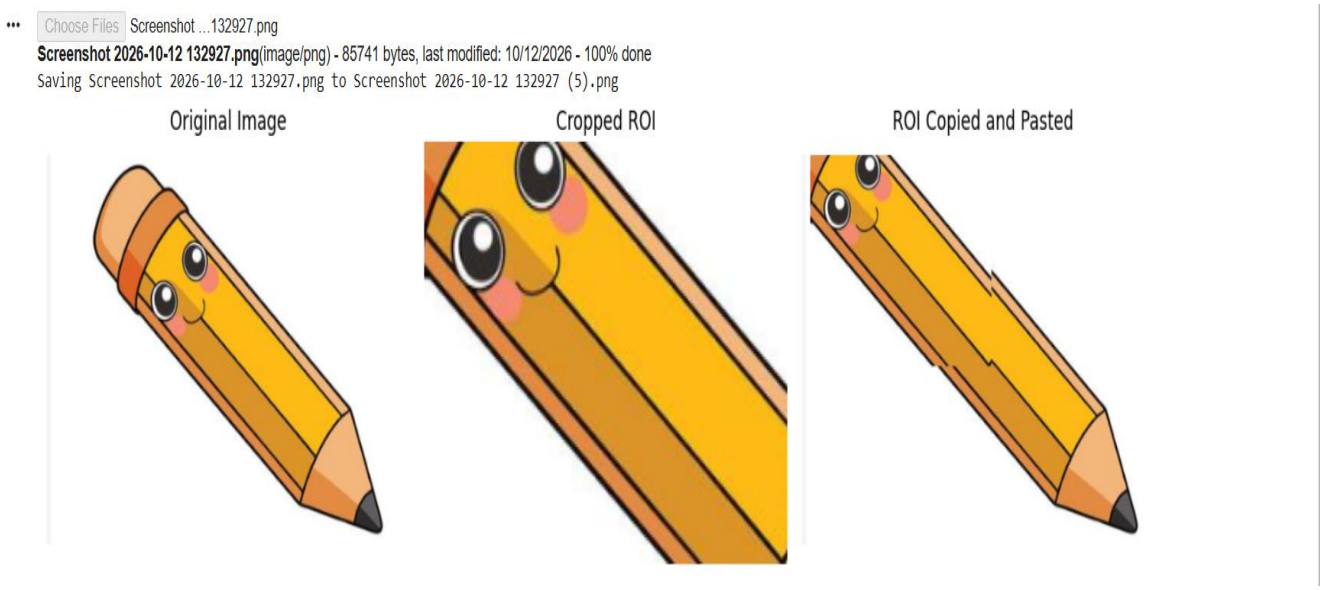
plt.tight_layout()
plt.show()
```

Expected Output

The output displays:

- Original Image.
- Cropped ROI.
- Image with the selected ROI copied and pasted at another location.

4. Output



The selected region of the original image is successfully cropped, copied and pasted at a different location.

5. Result

Thus, image cropping, copying and pasting operations using Region of Interest (ROI) were successfully implemented using Python and OpenCV.